

Cambridge (CIE) IGCSE Physics
Paper 4: Theory (Set D)
Extended

Saturday 26 April 2025

Morning (Time: 1 hour 15 minutes)

Total marks

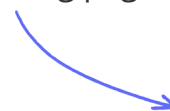
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Instructions

- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8N (acceleration of free fall = 9.8m/s^2).
- You should show all your working and use appropriate units.

Scan here to mark your mock exam

or visit the mock exams landing page for this course



1 (a) Define acceleration.

(1 mark)

(b) Fig. 1.1 shows the speed-time axes for the graph of the motion of a car.

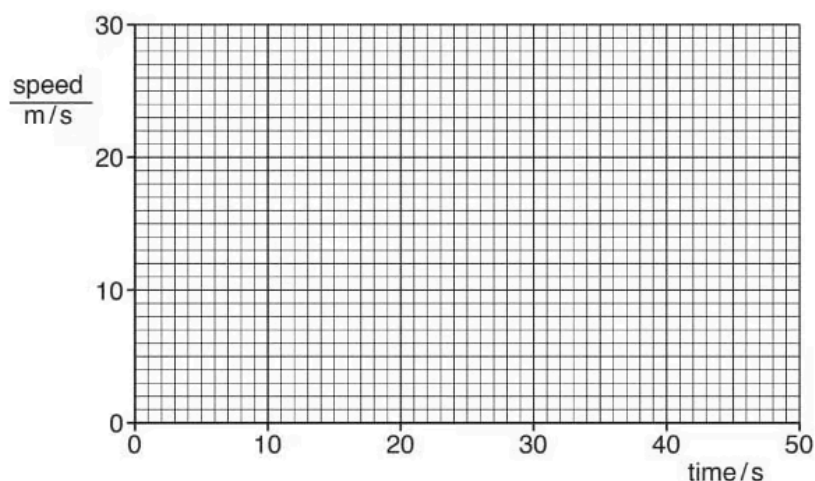


Fig. 1.1

(i) The car starts from rest.

From time = 0 to time = 15 s, the car has a constant acceleration to a speed of 28 m/s.

From time = 15 s to time = 32 s, the car has a constant speed of 28 m/s.

From time = 32 s, the car has a constant deceleration of 2.0 m/s^2 until it comes to rest.

On Fig. 1.1, draw the graph, using the space below for any calculations.

[5]

(ii) From time = 15 s to time = 32 s, the path of the car is part of a circle.

For this motion, state

1. the direction of the resultant force on the car,
2. what happens to the velocity of the car.

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(7 marks)

2 (a) A resultant force acts on an object at rest.

State the direction of the acceleration.

.....
(1 mark)

(b) A resultant force acts perpendicularly on a object traveling at a constant speed.

State the effect of the force on the object.

.....
(1 mark)

(c) A man with a mass of 70 kg steps into an elevator.

(i) State the value of the force exerted on the man by the elevator.

force = [1]

(ii) Calculate the force required to accelerate the man at 1.6 m/s^2 .

force = [1]

(iii) Explain why these values are different.

[2]
.....
.....
.....

(4 marks)

3 (a) Fig. 2.1 shows a fork-lift truck lifting a box.



Fig. 2.1

The electric motor that drives the lifting mechanism is powered by batteries.

Name the store that energy is usefully transferred from by the batteries.

(1 mark)

(b) The lifting mechanism raises a box of mass 32 kg through a vertical distance of 2.5 m in 5.4 s.

(i) Calculate the energy transferred to the gravitational potential energy store of the box.

energy =[2]

(ii) The efficiency of the lifting mechanism is 0.65 (65%).

Calculate the input power to the lifting mechanism.

input power =[3]

(5 marks)

- (c)** Batteries are recharged from a mains voltage supply that is generated in an oil-fired power station.

By comparison with a wind farm, state one advantage and one disadvantage of running a power station using oil.

(2 marks)

4 (a) Fig 2.1 shows liquid in a cylinder.

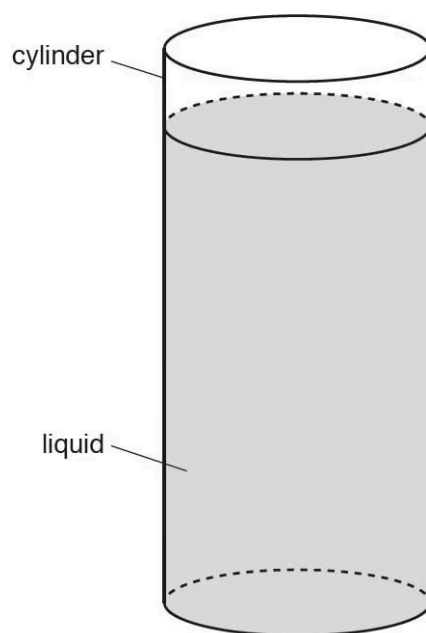


Fig. 2.1

Table 2.1 gives some data about the cylinder and the liquid.

Table 2.1

radius of cylinder	3.5 cm
weight of empty cylinder	2.5 N
depth of liquid	12.0 cm
density of liquid	900 kg/m^3

The cylinder containing liquid is placed on a digital balance that displays the mass in kg.

Calculate the reading shown on the balance.

readingkg

.....
.....

(4 marks)

(b) Fig. 2.2 shows a device called a manometer that measures the pressure of a gas.

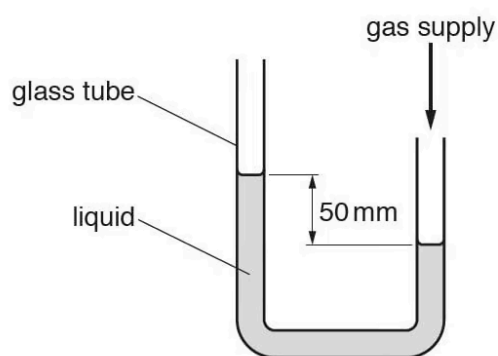


Fig. 2.2

(i) The pressure of the gas is 400 Pa greater than atmospheric pressure.

Calculate the density of the liquid.

density =[2]

(ii) With the gas supply connected, the top of the tube on the left of the device is sealed securely with a rubber stopper. The gas pressure is then increased.

State and explain what happens to the liquid in the manometer.

[2]

(4 marks)

5 (a) An object in the space between Earth and the Sun has an average temperature of 3 K.

(i) State how the intensity of radiation the object receives from the Sun at this point differs from the intensity at the Earth's surface.

[1]

(ii) Explain why the temperature of the object in space here is lower than if it was on Earth.

[2]

(3 marks)

(b) The average surface temperature of Venus is 475 °C.

The average surface temperature of Earth is 13 °C.

If Venus had an atmosphere the same as Earth's, it would have a maximum surface temperature of 35 °C.

If Venus had an orbital radius the same as Earth's, it would have a surface temperature of 227 °C.

Venus is ~30% closer to the Sun than the Earth is.

Referring to these data, explain why the surface temperature of Venus is much higher than Earth's.

(4 marks)

(c) The Earth's magnetic field arises from its iron-rich molten core, shown in Fig. 1.2.

Molten iron must be **in motion** to generate a magnetic field.

The inner core is very hot and solid. The outer core is also iron-rich and liquid.

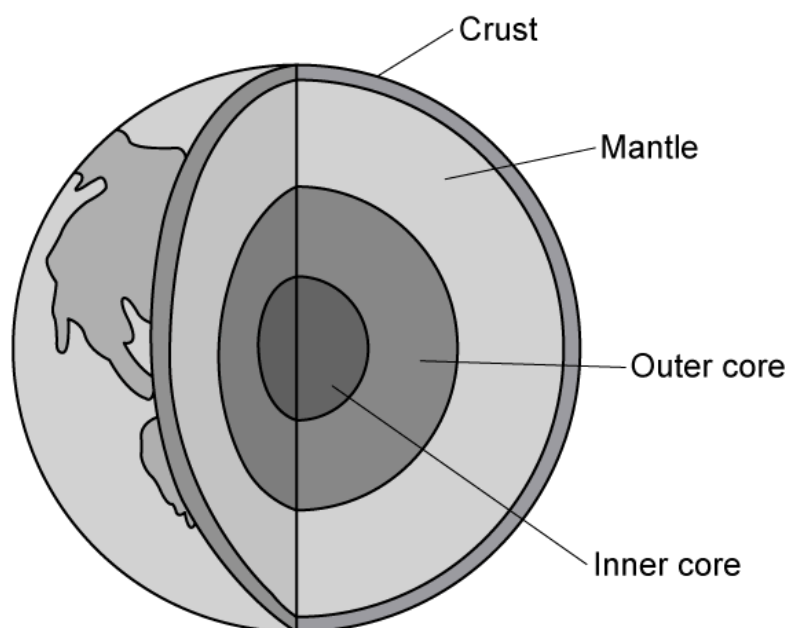


Fig. 1.2

Explain the origin of the Earth's magnetic field, referring to density in your answer.

(4 marks)

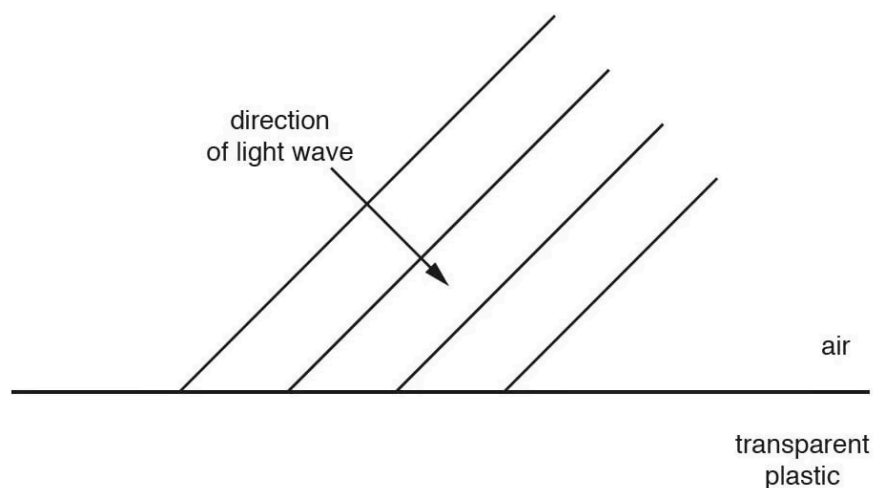
- 6 (a)** The speed of a light wave in air is $3.00 \times 10^8 \text{ m / s}$. The refractive index of water is 1.33.

Calculate the speed of the light wave in water.

speed =

(2 marks)

- (b)** Fig. 7.1 shows parallel wavefronts of a light wave in air incident on a boundary with a transparent plastic.



On Fig. 7.1,

- (i) draw the positions of the four refracted wavefronts in the plastic,

[3]

- (ii) draw an arrow to show the direction of travel of the refracted wave,

[1]

(iii) label the angle of refraction r of the light wave.

[1]

(5 marks)

- 7 (a)** Fig. 10.1 is a simplified top view of a flat coil. There is an alternating current (a.c.) in the coil.



Fig. 10.1

Describe the magnetic effect of this alternating current.

(2 marks)

- (b)** Fig. 10.2 shows a pan placed above the coil. The base of the pan is made of steel.

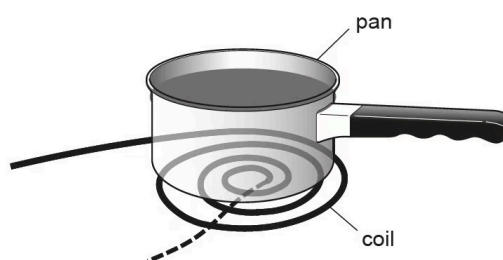


Fig. 10.2

State what quantity is induced in the base of the pan.

(1 mark)

- (c)** The pan contains water.

State and explain the effect of the quantity induced in part (b) on the temperature of the water in the pan.

(3 marks)

- 8 (a)** Fig. 10.1 shows a vacuum tube with a radioactive source. The radioactive source emits α -particles, β -particles and γ -rays. There is a very strong magnetic field between the N pole and the S pole of the magnet.

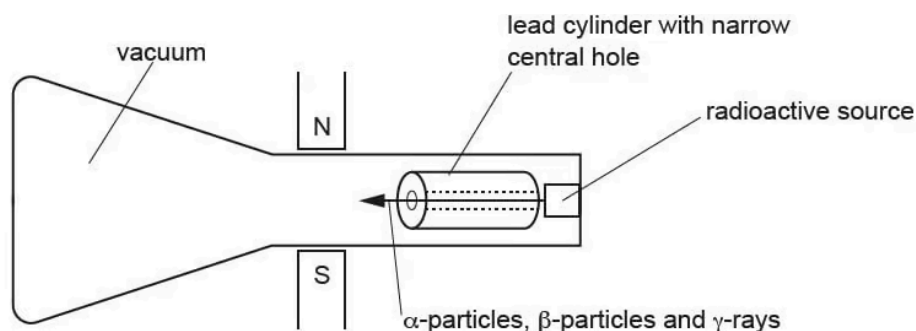


Fig. 10.1

The lead cylinder has a narrow central hole. State and explain the effect of the lead cylinder.

(2 marks)

- (b)** Describe the paths of the α -particles, β -particles and γ -rays as they pass through the magnetic field. Explain your answers.

(i) α -particles

[2]

(ii) β -particles

[2]

(iii) γ -rays

[2]

(6 marks)

9 (a) Fig. 10.1 represents a neutral atom of an isotope of element X.

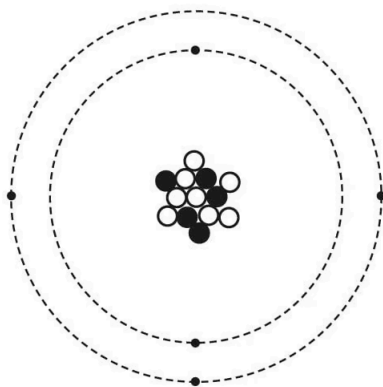


Fig. 10.1

State one similarity between this atom and a neutral atom of a different isotope of element X.

(1 mark)

(b) The isotope of element X is radioactive. It decays to form an isotope of element Y by emitting a β -particle.

(i) Using Fig. 10.1 deduce the nuclide notation for the isotope of Y produced by this decay.

nuclide notation: ${}^{A}_{Z}\text{Y}$ [3]

(ii) β -particles ionise the air they pass through less strongly than the same number of α -particles.

Suggest why this is so.

[3]

(6 marks)

10 (a) The 'big bang' theory is one theory explaining the origin of the Universe.

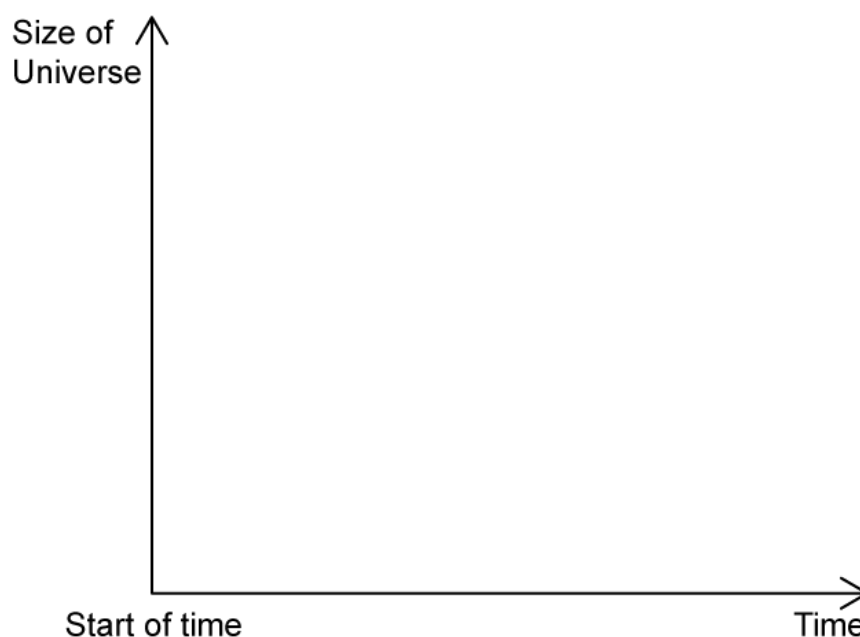


Fig 1.1

(i) Sketch the graph of the size of the Universe with time on Fig 1.1.

[2]

(ii) Explain how your graph in part (i) supports the Big Bang Theory.

[2]

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(4 marks)

(b) (i) Describe and explain the main properties of CMBR.

[2]

(ii) Explain how CMBR provides evidence for the evolution of the Universe.

[3]

(5 marks)

(c) Suggest why our understanding of the very earliest moments of the Universe is unreliable.

(2 marks)